

Experiment-12

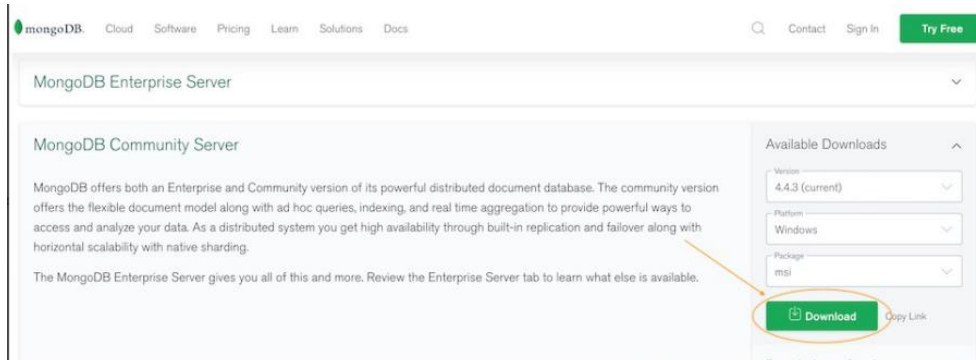
Aim: Install MongoDB on Windows

Procedure:

MongoDB is an open-source document-oriented database that is designed to store a large scale of data and also allows you to work with that data very efficiently. It is categorized under the NoSQL(Not only SQL) database because the storage and retrieval of data in MongoDB are not in the form of tables. This is the general introduction of MongoDB now we learn **how to install MongoDB on your Windows?**.

You can install MongoDB using two different methods one is using msi and another is using zip. Here, we will discuss how to install MongoDB using msi, so you need to follow each step carefully:

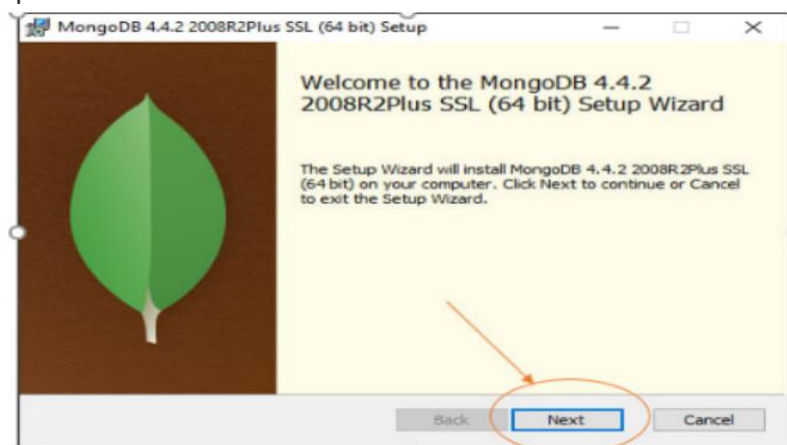
Step 1: Go to [MongoDB Download Center](#) to download MongoDB Community Server.



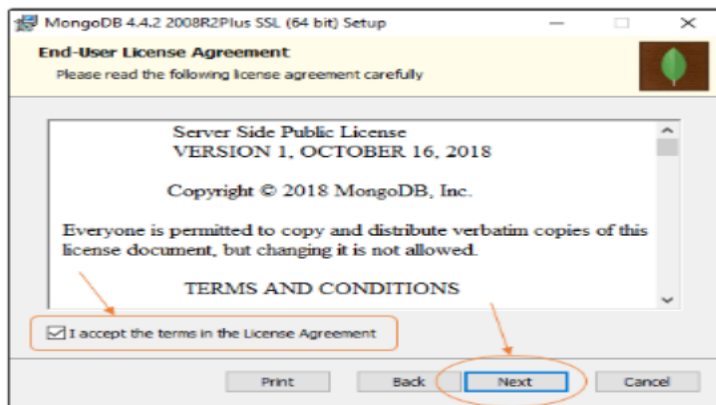
Here, You can select any version, Windows, and package according to your requirement. For Windows, we need to choose:

- Version: 4.2.2
- OS: WindowsOS
- Package: msi

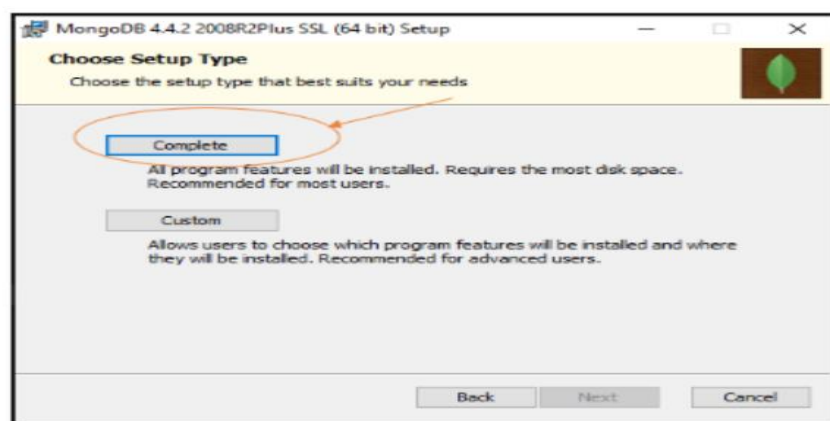
Step 2: When the download is complete open the msi file and click the *next button* in the startup screen:



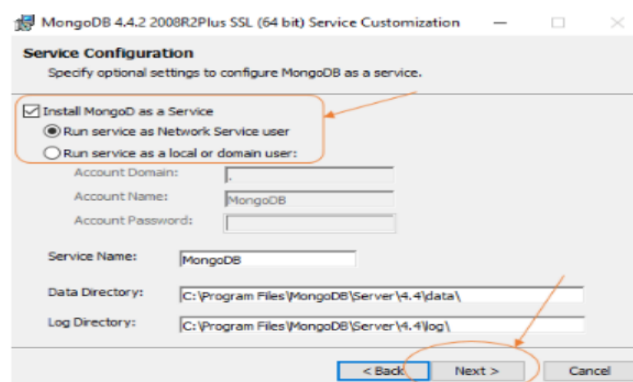
Step 3: Now accept the End-User License Agreement and click the next button:



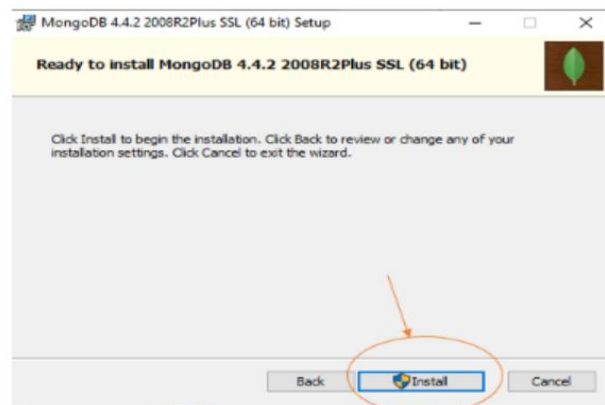
Step 4: Now select the *complete option* to install all the program features. Here, if you can want to install only selected program features and want to select the location of the installation, then use the *Custom option*:



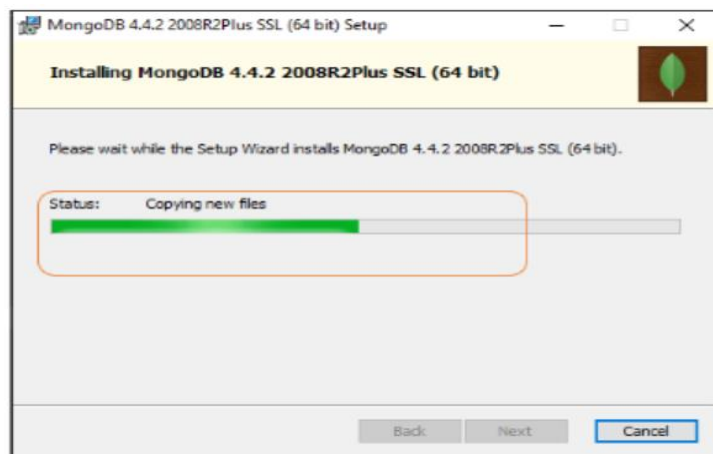
Step 5: Select “Run service as Network Service user” and copy the path of the data directory. Click Next:



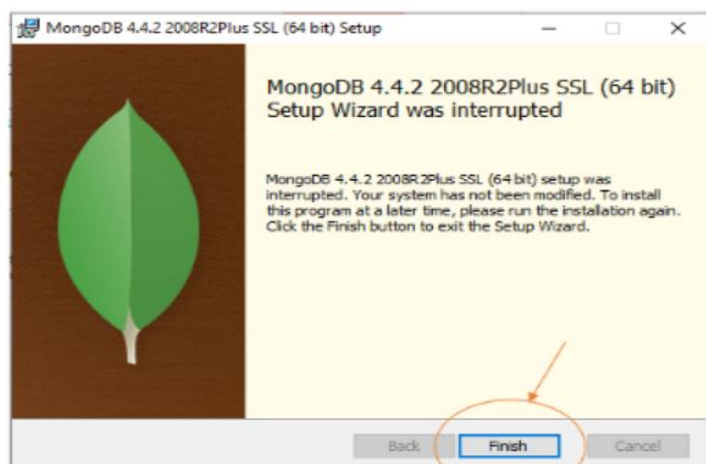
Step 6: Click the *Install* button to start the installation process:



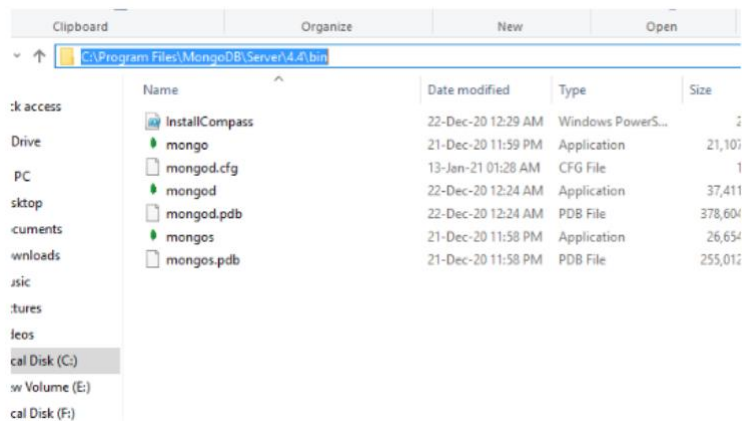
Step 7: After clicking on the install button installation of MongoDB begins:



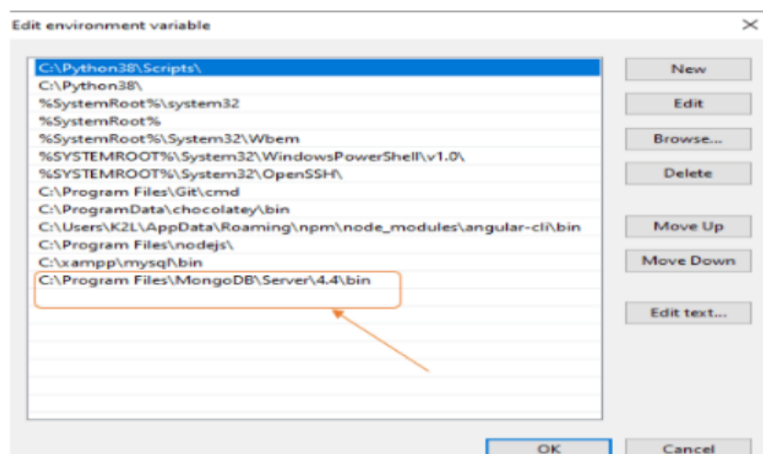
Step 8: Now click the *Finish* button to complete the installation process:



Step 9: Now we go to the location where MongoDB installed in step 5 in your system and copy the bin path:



Step 10: Now, to create an environment variable open system properties << Environment Variable << System variable << path << Edit Environment variable and paste the copied link to your environment system and click Ok:



Step 11: After setting the environment variable, we will run the MongoDB server, i.e. mongod. So, open the command prompt and run the following command:

```
mongod
```

When you run this command you will get an error i.e. *C:/data/db/ not found*.

Step 12: Now, Open C drive and create a folder named “data” inside this folder create another folder named “db”. After creating these folders. Again open the command prompt and run the following command: mongod

Now, this time the MongoDB server(i.e., mongod) will run successfully.

```

Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

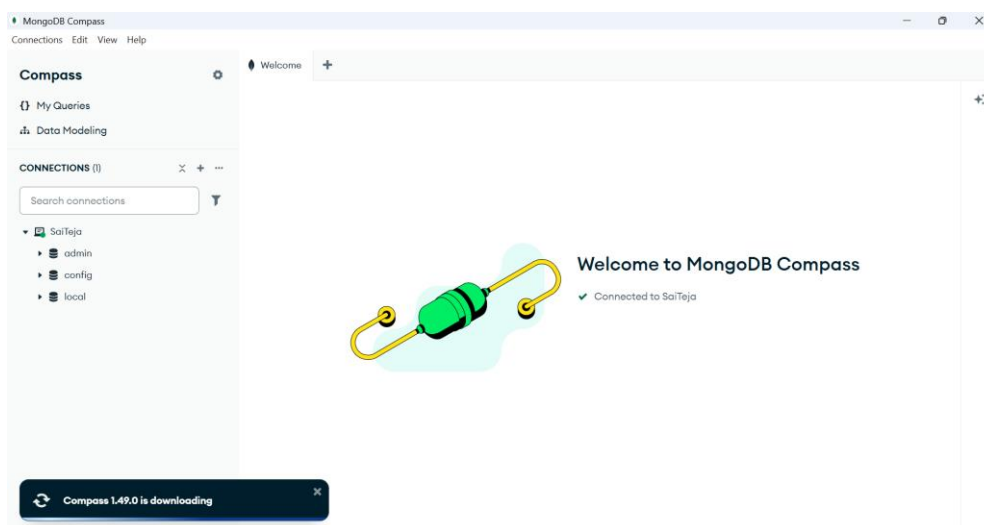
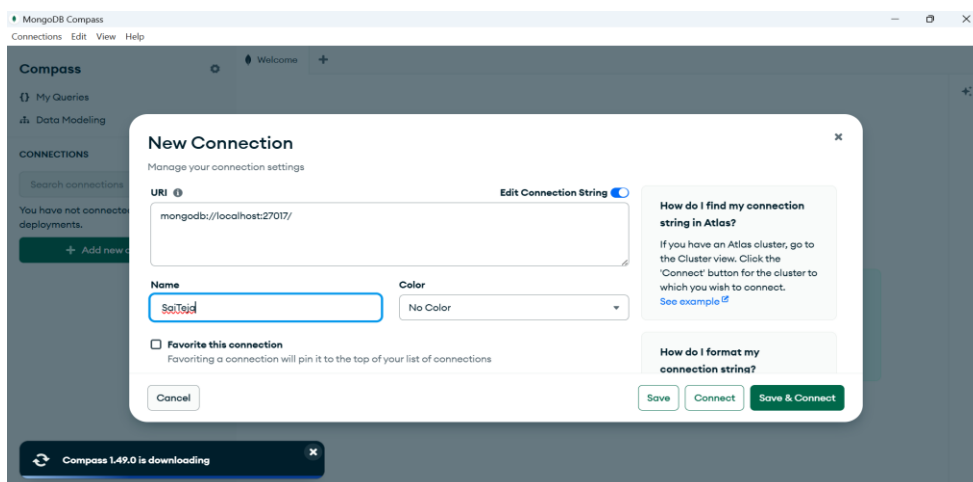
C:\Users\91974>mongod
{"t":{"$date":"2026-01-20T19:20:57.667+05:30"},"s":"I", "c":"-", "id":8991200, "ctx":"thread1","msg":"Shuffling initializers"
,"attr":{"seed":"2094033959"}}
{"t":{"$date":"2026-01-20T19:20:57.749+05:30"},"s":"I", "c":"CONTROL", "id":97374, "ctx":"thread1","msg":"Automatically disabling
TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS 1.0 specify --sslDisabledP
rotocols 'none'"}
{"t":{"$date":"2026-01-20T19:20:57.879+05:30"},"s":"I", "c":"NETWORK", "id":4915701, "ctx":"thread1","msg":"Initialized wire spec
ification", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWireVersion"
:0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient":true}}}}
{"t":{"$date":"2026-01-20T19:20:57.887+05:30"},"s":"I", "c":"CONTROL", "id":5945603, "ctx":"thread1","msg":"Multi threading initial
ized"}
{"t":{"$date":"2026-01-20T19:20:57.890+05:30"},"s":"I", "c":"CONTROL", "id":4615611, "ctx":"initandlisten","msg":"MongoDB starting"
,"attr":{"pid":15740,"port":27017,"dbPath":"/data/db","architecture":"64-bit","host":"LAPTOP-95G8BEHC"}}
{"t":{"$date":"2026-01-20T19:20:57.890+05:30"},"s":"I", "c":"CONTROL", "id":23398, "ctx":"initandlisten","msg":"Target operating
system minimum version", "attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-01-20T19:20:57.894+05:30"},"s":"I", "c":"CONTROL", "id":23403, "ctx":"initandlisten","msg":"Build Info", "attr
":{"buildInfo":{"version":"8.2.3","gitVersion":"36f41c9c30a2f13f834d033ba03c3463c891fb01","modules":[],"allocator":"tcmalloc-gperf",
"environment":{"distmod":"windows"}}}}
{"t":{"$date":"2026-01-20T19:20:57.895+05:30"},"s":"I", "c":"CONTROL", "id":51765, "ctx":"initandlisten","msg":"Operating System"
,"attr":{"os":{"name":"Microsoft Windows Workstation (build 22621)","version":"10.0 (build 22621)"}}}
{"t":{"$date":"2026-01-20T19:20:57.895+05:30"},"s":"I", "c":"CONTROL", "id":21951, "ctx":"initandlisten","msg":"Options set by co
mmand line", "attr":{"options":{}}}
{"t":{"$date":"2026-01-20T19:20:57.904+05:30"},"s":"I", "c":"NETWORK", "id":4648601, "ctx":"initandlisten","msg":"Implicit TCP Fast
Open unavailable. If TCP FastOpen is required, set at least one of the related parameters", "attr":{"relatedParameters":{"tcpFastOpenS
erver","tcpFastOpenClient","tcpFastOpenQueueSize"}}}}
{"t":{"$date":"2026-01-20T19:20:57.910+05:30"},"s":"E", "c":"CONTROL", "id":20557, "ctx":"initandlisten","msg":"DBException in in
itandlisten, terminating", "attr":{"error":{"message":"Data directory \\data\\db not found. Create the missing directory or spec
ify another path using (1) the --dbpath command line option, or (2) by adding the 'storage.dbPath' option in the configuration file."
}}}
{"t":{"$date":"2026-01-20T19:20:57.911+05:30"},"s":"I", "c":"REPL", "id":4784900, "ctx":"initandlisten","msg":"Stepping down the

```

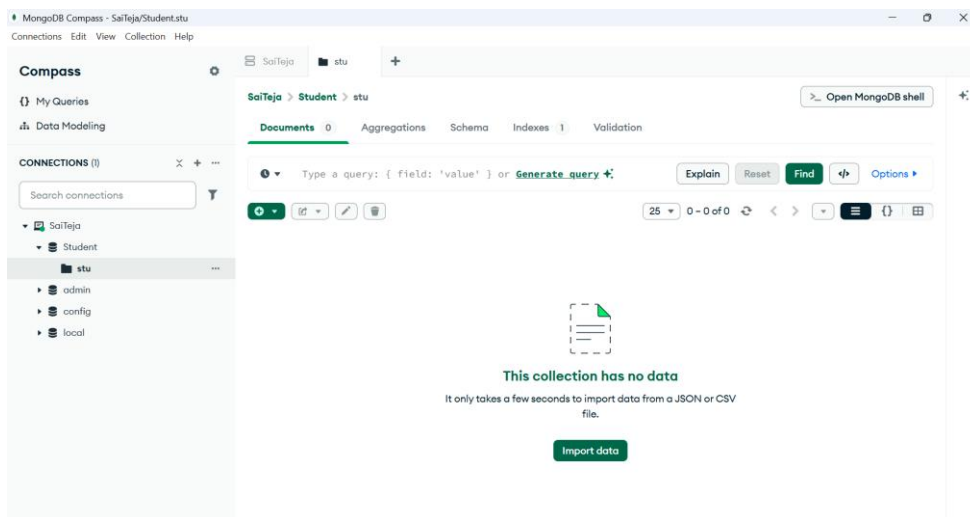
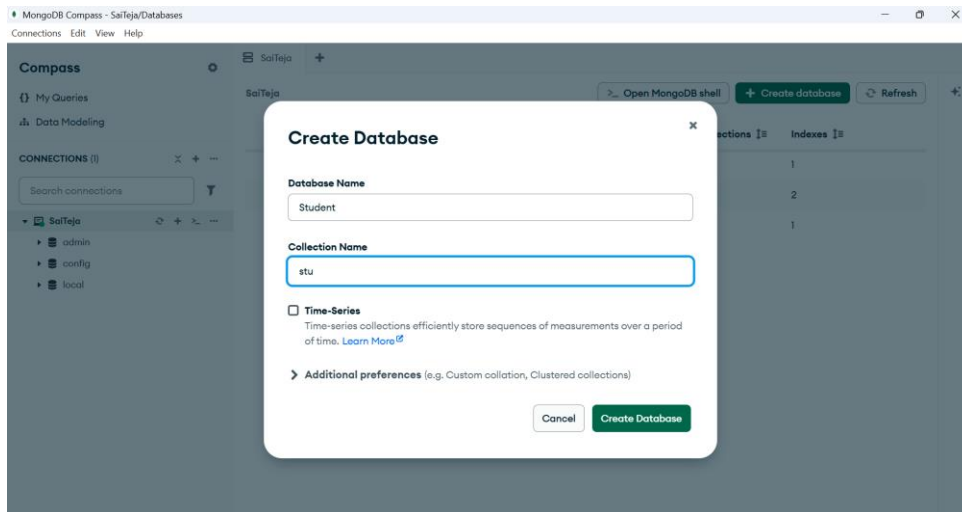
Run mongo Shell

Step 13: Now we are going to connect our server (mongod) with the mongo shell. So, keep that mongod window and open a new command prompt window and write **mongo**. Now, our mongo shell will successfully connect to the mongod.

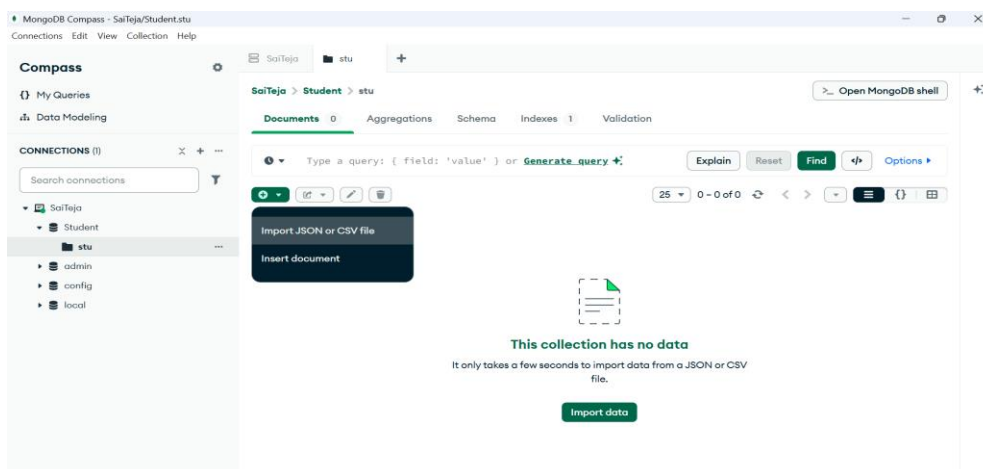
Add New Connection

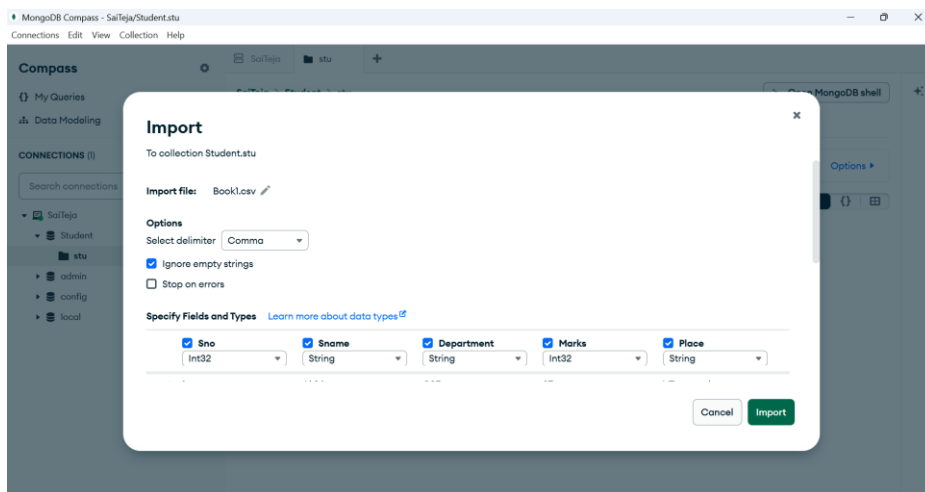
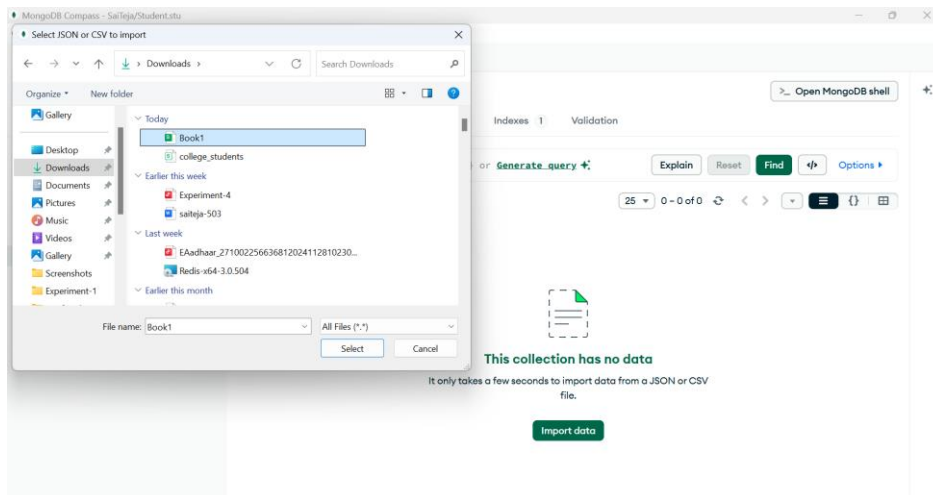


Create Database

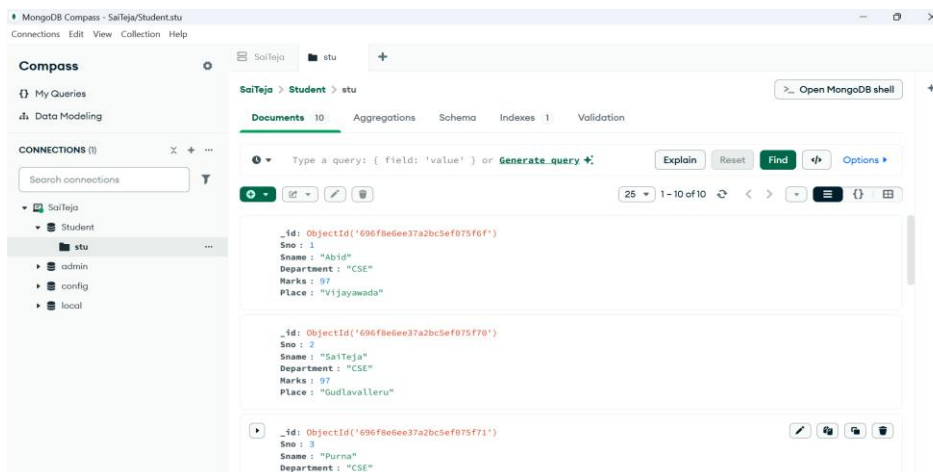


Create Document

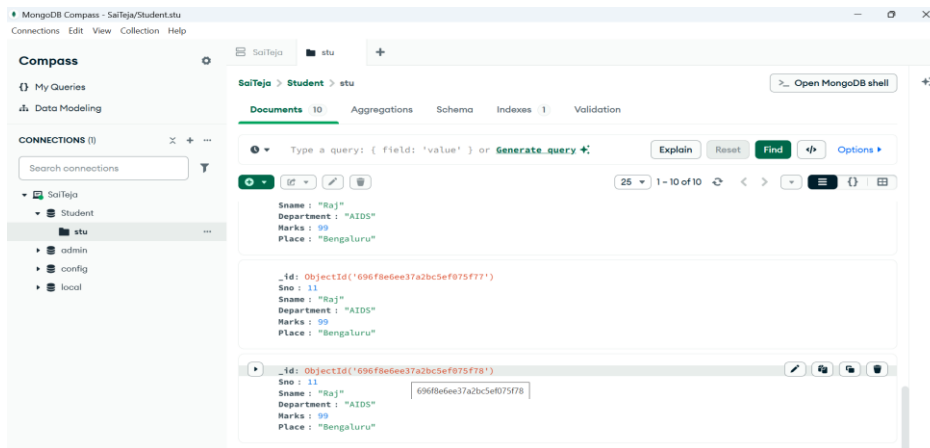




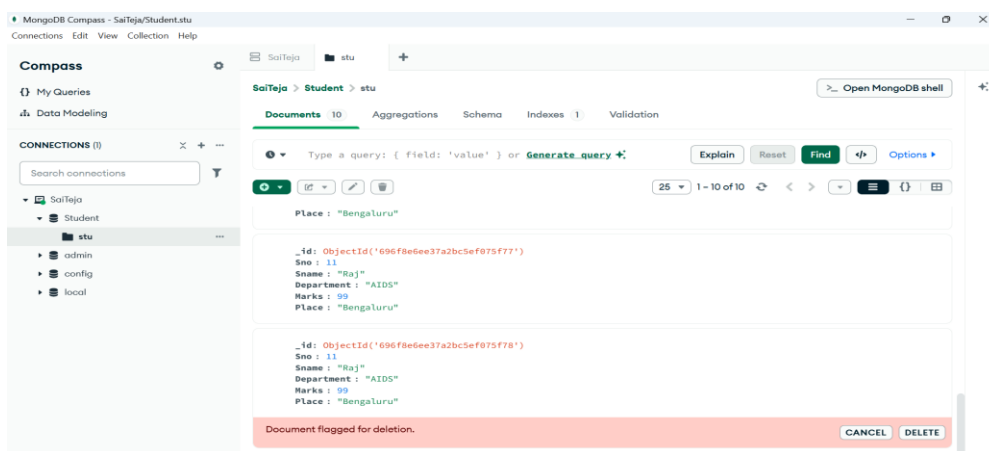
Retrieve the Data



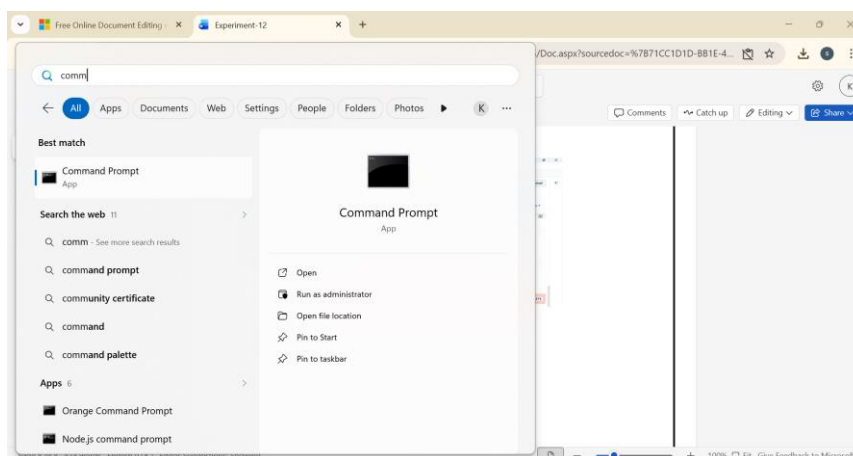
Update Document



Remove Document



Through Command Prompt



Microsoft Windows [Version 10.0.19045.6456]

(c) Microsoft Corporation. All rights reserved.


```

Microsoft Windows [Version 10.0.22621.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\91974>mongod
{"t":{"$date":"2026-01-20T19:56:11.212+05:30"},"s":"I",  "c":"-",          "id":8991200, "ctx":"thread1","msg":"Shuffling initializers"
  ,"attr":{"seed":2473981308}}
{"t":{"$date":"2026-01-20T19:56:11.226+05:30"},"s":"I",  "c":"CONTROL",  "id":97374,  "ctx":"thread1","msg":"Automatically disabling
  TLS 1.0 and TLS 1.1, to force-enable TLS 1.1 specify --sslDisabledProtocols 'TLS1_0'; to force-enable TLS 1.0 specify --sslDisabledP
  rotocols 'none'"}
{"t":{"$date":"2026-01-20T19:56:11.297+05:30"},"s":"I",  "c":"NETWORK",  "id":4915701, "ctx":"thread1","msg":"Initialized wire specifi
  cation", "attr":{"spec":{"incomingExternalClient":{"minWireVersion":0,"maxWireVersion":27},"incomingInternalClient":{"minWireVersion"
  :0,"maxWireVersion":27},"outgoing":{"minWireVersion":6,"maxWireVersion":27},"isInternalClient":true}}}
{"t":{"$date":"2026-01-20T19:56:11.298+05:30"},"s":"I",  "c":"CONTROL",  "id":5945603, "ctx":"thread1","msg":"Multi threading initial
  ized"}
{"t":{"$date":"2026-01-20T19:56:11.298+05:30"},"s":"I",  "c":"CONTROL",  "id":4615611, "ctx":"initandlisten","msg":"MongoDB starting"
  ,"attr":{"pid":4672,"port":27017,"dbPath":"/data/db","architecture":"64-bit","host":"LAPTOP-95G8BEHC"}}
{"t":{"$date":"2026-01-20T19:56:11.298+05:30"},"s":"I",  "c":"CONTROL",  "id":23398,  "ctx":"initandlisten","msg":"Target operating
  system minimum version", "attr":{"targetMinOS":"Windows 7/Windows Server 2008 R2"}}
{"t":{"$date":"2026-01-20T19:56:11.298+05:30"},"s":"I",  "c":"CONTROL",  "id":23403,  "ctx":"initandlisten","msg":"Build Info", "attr
 ":{"buildInfo":{"version":"8.2.3","gitVersion":"36f41c9c30a2f13f834d033ba03c3463c891fb01","modules":[],"allocator":"tcmalloc-gperf","
  environment":{"distmod":"windows"}}}}}
{"t":{"$date":"2026-01-20T19:56:11.299+05:30"},"s":"I",  "c":"CONTROL",  "id":51765,  "ctx":"initandlisten","msg":"Operating System"
  ,"attr":{"os":{"name":"Microsoft Windows Workstation (build 22621)","version":"10.0 (build 22621)"}}}}
{"t":{"$date":"2026-01-20T19:56:11.299+05:30"},"s":"I",  "c":"CONTROL",  "id":21951, "ctx":"initandlisten","msg":"Options set by co
  mmand line", "attr":{"options":{}}}
{"t":{"$date":"2026-01-20T19:56:11.300+05:30"},"s":"I",  "c":"NETWORK",  "id":4648601, "ctx":"initandlisten","msg":"Implicit TCP Fast
  Open unavailable. If TCP FastOpen is required, set at least one of the related parameters", "attr":{"relatedParameters":{"tcpFastOpenS
  ever","tcpFastOpenClient","tcpFastOpenQueueSize"}}}}
{"t":{"$date":"2026-01-20T19:56:11.302+05:30"},"s":"E",  "c":"CONTROL",  "id":20557,  "ctx":"initandlisten","msg":"DBException in i
  nitAndlisten, terminating", "attr":{"error":"NonExistentPath: Data directory \\data\\db not found. Create the missing directory or spec
  ify another path using (1) the --dbpath command line option, or (2) by adding the 'storage.dbPath' option in the configuration file."
  }}}
{"t":{"$date":"2026-01-20T19:56:11.303+05:30"},"s":"I",  "c":"REPL",    "id":4784900, "ctx":"initandlisten","msg":"Stepping down the

```

Current version of MongoDB

```

C:\Users\91974>mongod --version
db version v8.2.3
Build Info: {
  "version": "8.2.3",
  "gitVersion": "36f41c9c30a2f13f834d033ba03c3463c891fb01",
  "modules": [],
  "allocator": "tcmalloc-gperf",
  "environment": {
    "distmod": "windows"
  }
}
C:\Users\91974>

```

Go to Mongo Shell

```

test> { "exitCode":100}
mongosh
Current Mongosh Log ID: 696f909b433c99de3a1e2620
Connecting to:  mongodb://127.0.0.1:27017/?directConnection=true&serverSelectionTimeoutMS=2000&appName=mongosh+2.5.10
Using MongoDB:  8.2.3
Using Mongosh:  2.5.10
mongosh 2.6.0 is available for download: https://www.mongodb.com/try/download/shell

For mongosh info see: https://www.mongodb.com/docs/mongodb-shell/

-----
The server generated these startup warnings when booting
2026-01-20T19:06:43.213+05:30: Access control is not enabled for the database. Read and write access to data and configuration is
unrestricted
-----

test> |

```

VIEW ALL DATABASES

```

test> show dbs
Student  68.00 KiB
admin    40.00 KiB
config  108.00 KiB
local    72.00 KiB
test>

```

CREATE A NEW OR SWITCH DATABASE

test> use Student

switched to db Student

Student>

VIEW CURRENT DATABASE

Student> db

Student

Student>

DROP DATABASE

```
Student> use Student1
switched to db Student1
Student1> db.dropDatabase()
{ ok: 1, dropped: 'Student1' }
Student1>
```

CREATE DATABASE

```
Student1> use Student2

switched to db Student2

Student2>
```

CREATE COLLECTIONS

```
Student2> db.createCollection('stu');
{ ok: 1 }
Student2> db.createCollection('stu1');
{ ok: 1 }
Student2>
```

SHOW COLLECTIONS

```
Student2> show collections
stu
stu1
Student2>
```

CRUD OPERATIONS**INSERT-ONE-RECORD**

```
Student2> db.stu.insert({'sno':501,'sname':'Abid','dept':'CSE'})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('696f9423433c99de3a1e2622') }
}
Student2>
```

INSERT – MANY – RECORDS

```
Student2>db.stu.insertMany([{'sno':502,'sname':'Purna','dept':'CSE'},
{'sno':503,'sname':'Sai','dept':'CSE'}])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('696f947f433c99de3a1e2623'),
    '1': ObjectId('696f947f433c99de3a1e2624')
  }
}
Student2>
```

Retrieve Data

```
Student2> db.stu.find().pretty();
[
  {
    _id: ObjectId('696f9423433c99de3a1e2622'),
    sno: 501,
    sname: 'Abid',
    dept: 'CSE'
  },
  {
    _id: ObjectId('696f947f433c99de3a1e2623'),
    sno: 502,
    sname: 'Purna',
    dept: 'CSE'
  },
  {
    _id: ObjectId('696f947f433c99de3a1e2624'),
    sno: 503,
    sname: 'Sai',
    dept: 'CSE'
  }
]
Student2>
```

SERACH IN MONGODB

```
Student2> db.stu.find({sno:501})
[
  {
    _id: ObjectId('696f9423433c99de3a1e2622'),
    sno: 501,
    sname: 'Abid',
    dept: 'CSE'
  }
]
Student2>
```

LIMIT THE NO OF ROWS IN OUTPUT

```
Student2> db.stu.find().limit(2);
[
  {
    _id: ObjectId('696f9423433c99de3a1e2622'),
    sno: 501,
    sname: 'Abid',
    dept: 'CSE'
  },
  {
    _id: ObjectId('696f947f433c99de3a1e2623'),
    sno: 502,
    sname: 'Purna',
    dept: 'CSE'
  }
]
Student2>
```

COUNT THE NO OF ROWS IN THE OUTPUT

```
Student2> db.stu.find().count();
4
Student2>
```

UPDATE THE DOCUMENT

```
Student2>db.stu.updateOne({sno:501},{ $set:{'sno':5,'sname':'Ram','dept':'IT'}},{upsert:true}
)
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Student2>
```

```
Student2> db.stu.find().pretty();
[
  {
    _id: ObjectId('696f9416433c99de3a1e2621'),
    sno: 5,
    sname: 'Ram',
    dept: 'IT'
  },
]
```

```
{
  _id: ObjectId('696f9423433c99de3a1e2622'),
  sno: 501,
  sname: 'Abid',
  dept: 'CSE'
},
{
  _id: ObjectId('696f947f433c99de3a1e2623'),
  sno: 502,
  sname: 'Purna',
  dept: 'CSE'
},
{
  _id: ObjectId('696f947f433c99de3a1e2624'),
  sno: 503,
  sname: 'Sai',
  dept: 'CSE'
}
]
```

Student2>

REMOVE DOCUMENT

Student2> db.stu.remove({sno:5})

DeprecationWarning: Collection.remove() is deprecated. Use deleteOne, deleteMany, findOneAndDelete, or bulkWrite.

```
{ acknowledged: true, deletedCount: 1 }
```

Student2>

Student2> db.stu.find().pretty();

```
[
  {
    _id: ObjectId('696f947f433c99de3a1e2623'),
    sno: 502,
    sname: 'Purna',
    dept: 'CSE'
  },
  {
    _id: ObjectId('696f947f433c99de3a1e2624'),
    sno: 503,
    sname: 'Sai',
    dept: 'CSE'
  }
]
```

